Sounak Kar

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Experience

Postdoc Oct'23 – Present
QuTech, TU Delft *Delft, The Netherlands*

- Research topic: Modelling and performance analysis of quantum networks.

Postdoc Dec'21 – July'23
EPFL *Lausanne, Switzerland*

- Research topic: Modelling and performance analysis of quantum networks.
- Teaching: TA for TCP/IP Networking and Smart Grid Technologies.

Research Associate Apr'17 – Dec'21
DFG MAKI *Darmstadt, Germany*

- Research topic: Modelling and performance analysis of classical networks.
- Teaching: TA for Communication Networks I and Communication Networks IV at TU Darmstadt.

Associate Jul'14 – Mar'17
Nomura Holdings *Mumbai, India*

- Worked on models for collaterals.

Manager Jun'10 – Jul'14
ICICI Bank *Mumbai, India*

- Built and validated models for credit and market risk assessment.

Education

Technische Universität Darmstadt Darmstadt, Germany
Ph.D., Electrical Engineering *Apr'17 – Oct'21*

Indian Statistical Institute Kolkata, India
Master of Statistics *Jul'08 – May'10*

Indian Statistical Institute Kolkata, India
Bachelor of Statistics (Hons.) *Jul'05 – May'08*

Publications and Preprints

Preprints

- Maiti, S., Avis, G., **Kar, S.** and Wehner, S., 2026. *Requirements for Teleportation in an Intercity Quantum Network*. arxiv preprint, https://arxiv.org/pdf/2602.04869.
- **Kar, S.**, Le Boudec, JY., 2023. *An Analysis of the Completion Time of the BB84 Protocol*. arxiv preprint, https://arxiv.org/abs/2304.10218 (workshop version at IEEE ISPASS 2023).
- **Kar, S.**, Alt, B., Koeppl, H. and Rizk, A., 2022. *Optimal Decision Making in Active Queue Management*. arxiv preprint, https://arxiv.org/abs/2202.10352.

Journal papers

- Iñesta, Á.G., Davies, B., **Kar, S.**, and Wehner, S., 2025. *Entanglement Buffering with Multiple Quantum Memories*. Accepted in npj Quantum Information.
- **Kar, S.**, Wehner, S., 2024. *Convexification of the Quantum Network Utility Maximisation Problem*. IEEE Transactions on Quantum Engineering.
- **Kar, S.**, Rehrmann, R., Mukhopadhyay, A., Alt, B., Ciucu, F., Koeppl, H., Binnig, C. and Rizk, A., 2020. *On the Throughput Optimization in Large-Scale Batch-Processing Systems*. Performance Evaluation, 144, pp.102-142.
- KhudaBukhsh, W.R., **Kar, S.**, Alt, B., Rizk, A. and Koeppl, H., 2020.*Generalized Cost-based Job Scheduling in Very Large Heterogeneous Cluster Systems*. IEEE Transactions on Parallel and Distributed Systems, 31(11), pp.2594-2604.
- KhudaBukhsh, W.R., **Kar, S.**, Rizk, A. and Koeppl, H., 2019. *Provisioning and Performance Evaluation of Parallel Systems with Output Synchronization*. ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS), 4(1), pp.1-31.

- Alt, B., Weckesser, M., Becker, C., Hollick, M., **Kar, S.**, Klein, A., Klose, R., Kluge, R., Koeppl, H., Koldehofe, B. and KhudaBukhsh, W.R., 2019. *Transitions: A Protocol-independent View of the Future Internet.* Proceedings of the IEEE, 107(4), pp.835-846.

Conference papers

- **Kar, S.**, Mukhopadhyay, A., 2026. *On Utility-optimal Entanglement Routing in Quantum Networks.* Accepted in International Conference on Quantum Communications, Networking, and Computing (QCNC).
- Hark, R., Ghanmi, M., **Kar, S.**, Richerzhagen, N., Rizk, A. and Steinmetz, R., 2018. *Representative Measurement Point Selection to Monitor Software-defined Networks.* In IEEE Conference on Local Computer Networks (LCN) (pp. 511-518). IEEE.
- KhudaBukhsh, W.R., Alt, B., **Kar, S.**, Rizk, A. and Koeppl, H., 2018. *Collaborative Uploading in Heterogeneous Networks: Optimal and Adaptive Strategies.* In IEEE INFOCOM- IEEE Conference on Computer Communications (pp. 1-9). IEEE.
- **Kar, S.**, Hark, R., Rizk, A. and Steinmetz, R., 2018, February. *Towards Optimal Placement of Monitoring Units in Time-Varying Networks Under Centralized Control.* In International Conference on Measurement, Modelling and Evaluation of Computing Systems (pp. 99-112). Springer, Cham.

Short papers

- **Kar, S.** and Rizk, A., 2020, June. *A Delicate Union of Batching and Parallelization Models in Distributed Computing and Communication.* In IFIP Networking Conference (Networking) (pp. 534-538). IEEE.
- **Kar, S.**, Rizk, A. and Fidler, M., 2018, September. *Multi-interface Communication: Interface Selection under Statistical Performance Constraints.* In International Teletraffic Congress (ITC 30) (pp. 7-12). IEEE.

Talks

- LORIA, Online, Jan'26.
- *Tutorial* on Quantum Communication and Networking, IFIP Performance, Amsterdam, Nov'25.
- Quantum Workshop, National Conference in Communication, IIT Madras, Chennai, Feb'24 (*invited*).
- QuTech, TU Delft, Delft, Jun'23.
- ISPASS, Workshop on Performance Analysis of Quantum Computer Systems, Raleigh, Apr'23.
- CNI Seminar Series, Indian Institute of Science, Bangalore, Jan'23.
- EPFL IC, Online, Jun'21.
- MAKI Scientific Advisory Board meeting, TU Darmstadt, Darmstadt, Mar'21.
- IFIP Performance, Online, Nov'20.
- ICICI Bank, Business Intelligence Unit, Online, Jul'20.
- IFIP Networking, Network Modeling workshop, Online, Jun'20.
- ITC, NetCal workshop, Vienna, Sep'18.
- MMB, Erlangen, Feb'18.

Professional Duties

- Reviewing activities: Theory of Quantum Computation, Communication and Cryptography (TQC); IEEE/ACM Transactions on Networking (ToN); Computer Communication; Computer Networks.
- Organization: Local organization team (EPFL) of Workshop on Network Calculus (Sep'22).

Other Skills

Programming: Matlab, R, C.
Languages: Bengali (Native), Hindi (Fluent), English (Fluent), German (Basic).

References

Ralf Steinmetz,
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TU Darmstadt,
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Jean-Yves Le Boudec,
Professor Emeritus, School of Computer and Communication Sciences,
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Stephanie Wehner,
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TU Delft,
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